Systematic Benchmarking of AI-based Scoring Function Against MM-PBSA for RNA–Ligand Binding Free Energy Estimation

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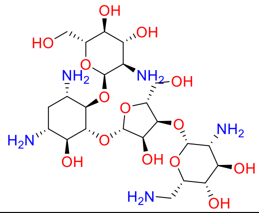
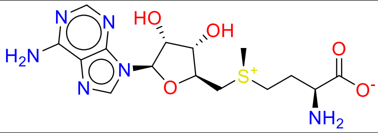
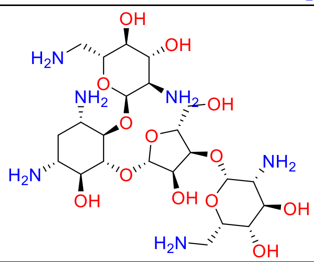
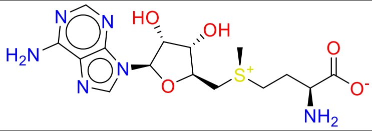
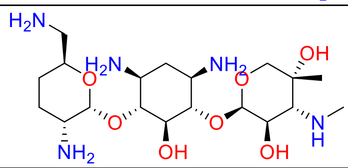
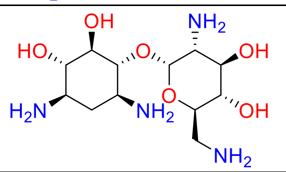
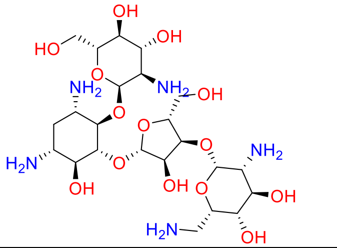
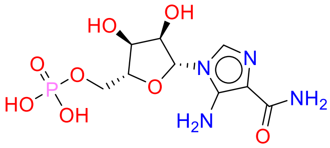
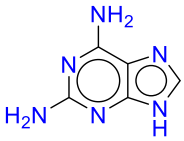
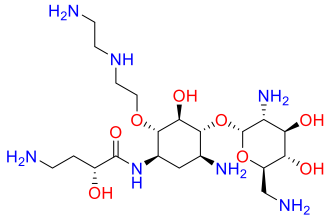
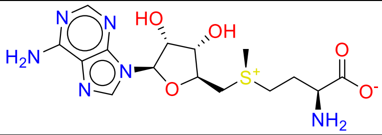
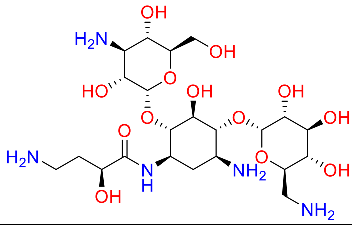
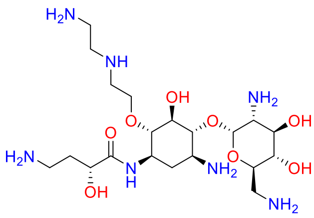
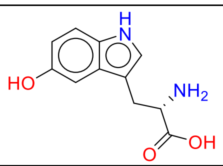
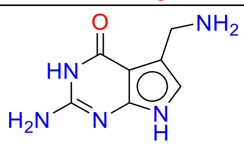
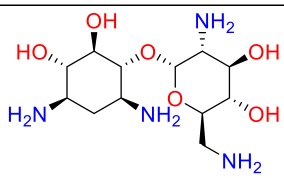
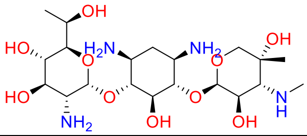
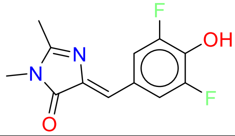
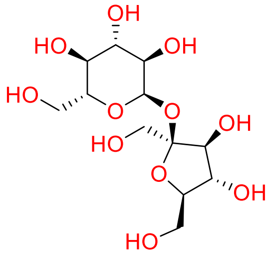
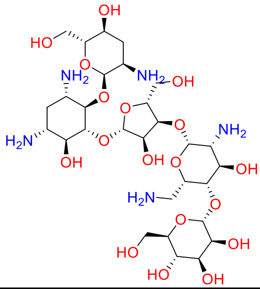
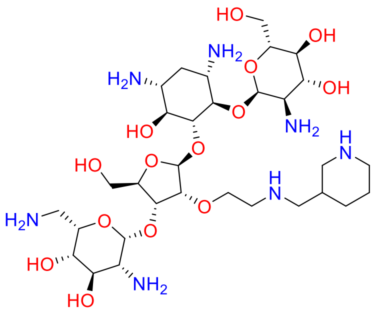
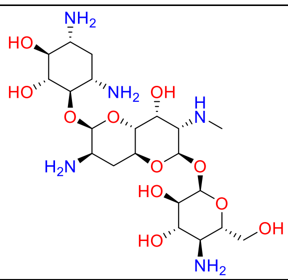
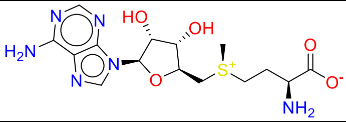
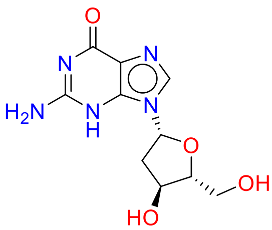
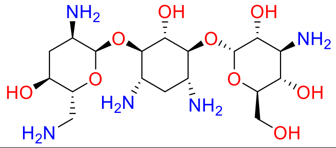
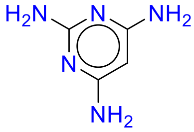
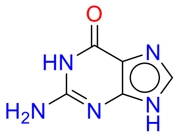
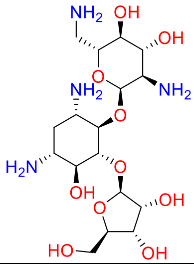
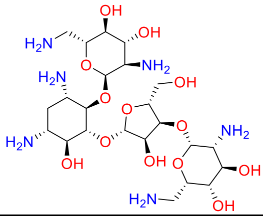
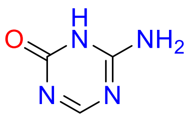
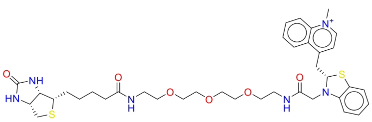
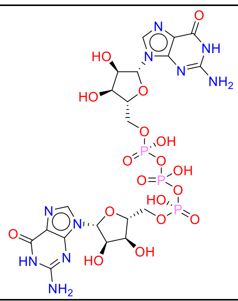
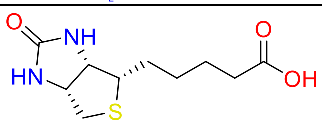
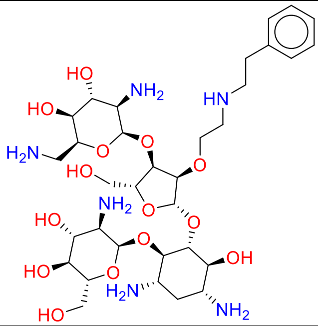
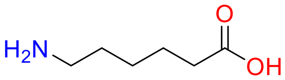
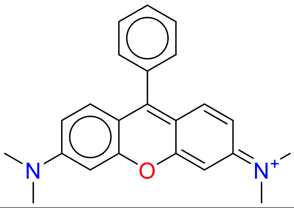
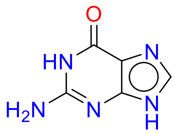
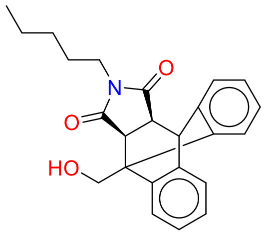
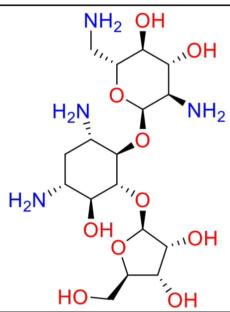
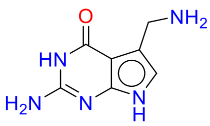
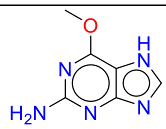
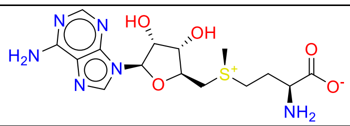
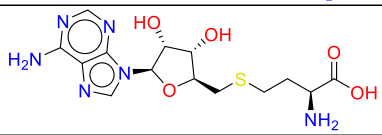
Supplementary Table S2. Structures of representative small-molecule ligands from RNA–ligand complexes with their predicted binding free energies using MM-PBSA and RNALig and comparison with experimental binding affinities.

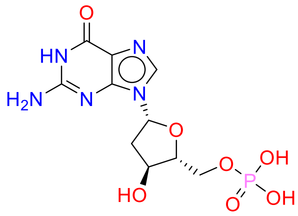
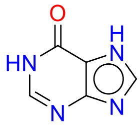
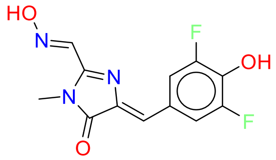
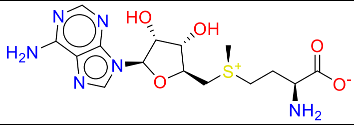
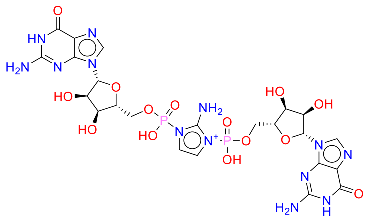
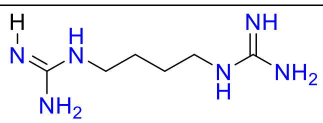
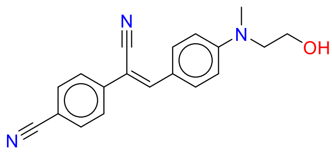
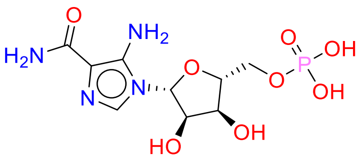
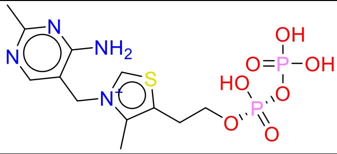
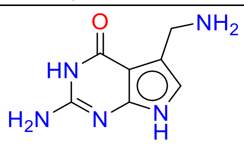
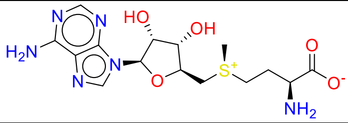
PDB ID	Ligand name	Structure	ΔG_MM-PBSA (kcal/mol)	ΔG_RNALig _frames (kcal/mol)	ΔG_Exp (kcal/mol)
1J7T	Paromomycin		-12.05	-8.12	-7.71
4OQU	S-adenosylmethionine		-16.97	-7.88	-8.71
2ET4	Neomycin		-23.71	-9.92	-10.85
3E5C	S-adenosylmethionine		-15.09	-11.82	-12.6
2ET3	Gentamicin c1a		-16.52	-11.41	-11.63
2ET8	(1R,2R,3S,4R,6S)-4,6-diamino-2,3-dihydroxycyclohexyl 2,6-diamino-2,6-dideoxy-alpha-D-glucopyranoside		-15.67	-8.09	-8.59

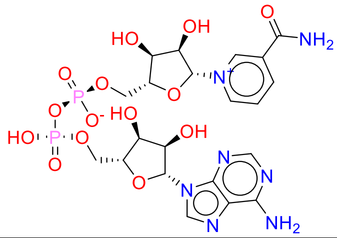
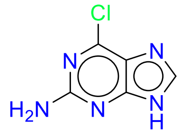
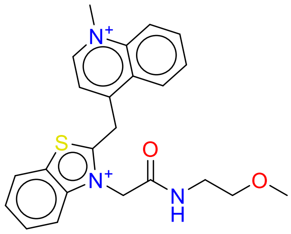
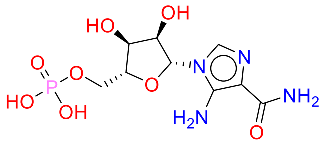
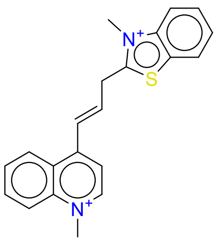
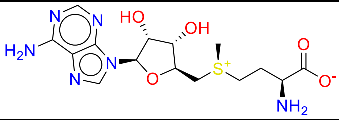
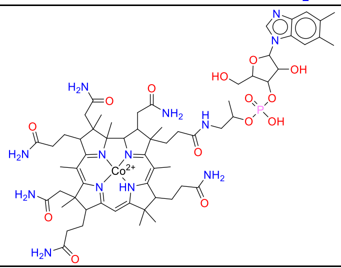
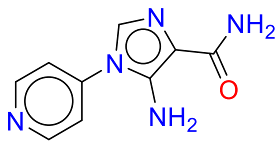
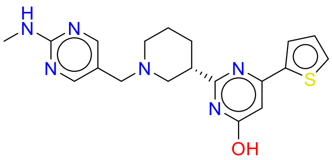
2O3W	Paromomycin		-20.56	-8.43	-8.21
4XWF	Aminoimidazole 4-carboxamide ribonucleotide		-16.05	-8.51	-8.21
2B57	9h-purine-2,6-diamine		-25.14	-10.06	-10.59
2F4T	(2R)-4-amino-N-[(1R,2S,3R,4R,5S)-5-amino-4-[(2R,3R,4R,5S,6R)-3-amino-6-(aminomethyl)-4,5-dihydroxy-oxan-2-yl]oxy-2-[2-(2-aminoethylamino)ethoxy]-3-hydroxy-cyclohexyl]-2-hydroxy-butanamide		-15.92	-7.19	-7.71
4L81	S-adenosylmethionine		-15.22	-6.74	-7.08
4P20	Amikacin		-20.74	-9.19	-9.95
2F4U	(2R)-4-amino-N-[(1R,2S,3R,4R,5S)-5-amino-4-[(2R,3R,4R,5S,6R)-3-amino-6-(aminomethyl)-4,5-dihydroxy-oxan-2-yl]oxy-2-[2-(2-aminoethylamino)ethoxy]-3-hydroxy-cyclohexyl]-2-hydroxy-butanamide		-18.69	-6.90	-8.02
5KPY	5-hydroxy-L-tryptophan		-19.71	-7.19	-7.38
4RZD	7-deaza-7-aminomethyl-guanine		-19.65	-8.53	-8.93
2F4S	Neomycin a		-13.55	-9.42	-9.21

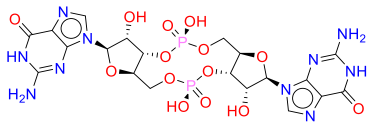
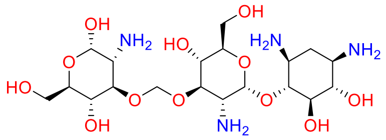
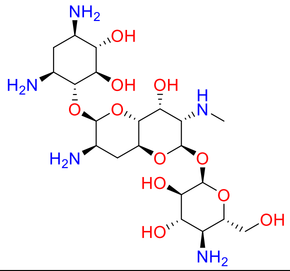
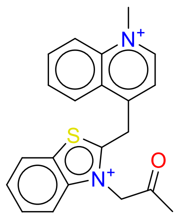
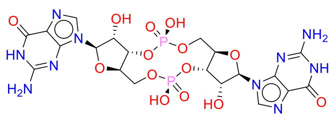
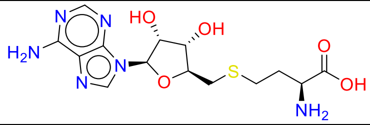
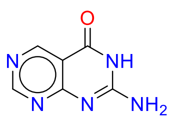
1MWL	Geneticin		-18.22	-8.87	-8.95
4TS2	(5Z)-5-(3,5-difluoro-4-hydroxybenzylidene)-2,3-dimethyl-3,5-dihydro-4H-imidazol-4-one		-28.85	-8.55	-8.56
4TS0	Sucrose		-29.51	-8.58	-8.56
2ESJ	Lividomycin a		-23.42	-9.70	-10.62
2BE0	(2s,3s,4r,5r,6r)-5-amino-2-(aminomethyl)-6-((2r,3r,4r,5s)-5-((1r,2r,3s,5r,6s)-3,5-diamino-2-((2s,3r,4r,5s,6r)-3-amino-4 ,5-dihydroxy-6-(hydroxymethyl)-tetrahydro-2h-pyran-2-yloxy)-6-hydroxycyclohexyloxy)-2-(hydroxymethyl)-4-(2-((r)-piperidi n-3-ylmethylamino)ethoxy)-tetrahydrofuran-3-yloxy)-tetrahydro-2h-pyran-3,4-diol		-18.54	-9.57	-8.83
1YRJ	Apramycin		-11.05	-5.34	-7.86
2GIS	S-adenosylmethionine		-29.08	-10.77	-10.91
3DS7	2'-deoxy-guanosine		-22.14	-6.41	-6.01

1LC4	Tobramycin		-12.74	-7.98	-7.11
2G9C	Pyrimidine-2,4,6-triamine		-22.33	-6.91	-7.22
6UBU	Guanine		-16.95	-10.75	-11.32
2ET5	Ribostamycin		-16.89	-8.44	-7.58
1I9V	Neomycin		-21.55	-5.31	-8.72
3LA5	5-azacytosine		-17.46	-8.01	-8.18
5V3F	4-{[(2S)-3-{2,16-dioxo-20-[(3as,4S,6ar)-2-oxohexahydro-1H-thieno[3,4-d]imidazol-4-yl]-6,9,12-trioxa-3,15-diazaicosan-1-yl}-2,3-dihydro-1,3-benzothiazol-2-yl]methyl}-1-methylquinolin-1-ium		-24.51	-12.62	-11.62
5UEE	Diguanosine-5'-triphosphate		-20.42	-8.06	-8.66
1F27	Biotin		-16.31	-9.86	-8.18

3S4P	(1R,2R,3S,4R,6S)-4,6-diamino-2-{[3-O-(2,6-diamino-2,6-dideoxy-beta-L-idopyranosyl)-2-O-{2-[(2-phenylethyl)amino]ethyl}-beta-D-ribofuranosyl]oxy}-3-hydroxycyclohexyl 2-amino-2-deoxy-alpha-D-glucopyranoside		-19.67	-9.11	-8.18
4ERJ	Aminocaproic acid		-30.24	-5.59	-6.42
1F1T	N,n'-tetramethyl-rosamine		-18.88	-8.81	-8.79
1Y27	Guanine		-9.81	-5.31	-4.97
1YKV	(3as,9as)-2-pentyl-4-hydroxymethyl-3a,4,9,9a-tetrahydro-4,9[1',2']-benzeno-1h-benz[f]isoindole-1,3(2h)-dione		-27.41	-7.15	-6.82
2FCZ	Ribostamycin		-14.05	-6.88	-6.79
3FU2	7-deaza-7-aminomethyl-guanine		-15.29	-9.89	-10.03
3GES	6-O-methylguanine		-20.87	-7.41	-7.63
3GX7	S-adenosylmethionine		-12.38	-6.99	-7.15
3NPQ	S-adenosyl-l-homocysteine		-12.02	-10.08	-10.26

3SLM	2'-deoxyguanosine-5'-monophosphate		-18.32	-5.11	-4.47
4FE5	Hypoxanthine		-16.84	-8.62	-8.37
5BJP	(5Z)-5-[(3,5-difluoro-4-hydroxyphenyl)methylidene]-2-[(E)-(hydroxyimino)methyl]-3-methyl-3,5-dihydro-4H-imidazol-4-one		-14.06	-7.66	-7.75
5FK4	S-adenosylmethionine		-11.92	-8.41	-8.62
6C8E	2-amino-1-[(R)-{[(2R,3S,4R,5R)-5-(2-amino-6-oxo-1,6-dihydro-9H-purin-9-yl)-3,4-dihydroxyoxolan-2-yl]methoxy}(hydroxy)phosphoryl]-3-[(S)-{[(2R,3S,4R,5R)-5-(2-amino-6-oxo-1,6-dihydro-9H-purin-9-yl)-3,4-dihydroxyoxolan-2-yl]methoxy}(hydroxy)phosphoryl]-1H-imidazol-3-ium		-12.09	-9.11	-9.22
6HBT	1-(4-carbamimidamidobutyl)guanidine		-20.21	-7.52	-7.75
7EOI	4-[(~{Z})-1-cyano-2-[4-[2-hydroxyethyl(methyl)amino]phenyl]ethenyl]benzenecarbonitrile		-18.06	-9.65	-9.49
4ZNP	Aminoimidazole 4-carboxamide ribonucleotide		-15.22	-5.26	-5.91
2GDI	Thiamine diphosphate		-17.75	-8.23	-8.48
3Q50	7-deaza-7-aminomethyl-guanine		-17.26	-9.11	-8.88
2QWY	S-adenosylmethionine		-15.82	-8.96	-9.26

7D7W	Nicotinamide-adenine-dinucleotide		-16.68	-4.85	-5.31
3FO4	6-chloroguanine		-11.24	-9.44	-9.79
6C63	4-[(3-{2-[(2-methoxyethyl)amino]-2-oxoethyl}-1,3-benzothiazol-3-ium-2-yl)methyl]-1-methylquinolin-1-ium		-15.68	-9.71	-10.03
5BTP	Aminoimidazole 4-carboxamide ribonucleotide		-21.39	-6.92	-7.61
6C64	1-methyl-4-[(1E)-3-(3-methyl-1,3-benzothiazol-3-ium-2-yl)prop-1-en-1-yl]quinolin-1-ium		-16.71	-12.74	-13.25
2YGH	S-adenosylmethionine		-21.1	-8.55	-8.87
1DDY	Cobalamin		-13.23	-7.02	-7.62
6WZR	5-amino-1-(pyridin-4-yl)-1H-imidazole-4-carboxamide		-19.08	-9.92	-10.98
5C45	2-[(3~{S})-1-[[2-(methylamino)pyrimidin-5-yl]methyl]piperidin-3-yl]-6-thiophen-2-yl-pyrimidin-4-ol		-22.02	-10.85	-11.15

3MXH	Cyclic diguanosine monophosphate		-39.21	-11.02	-11.42
2O3V	(2S,3R,4R,5S,6R)-3-amino-4-[[(2S,3R,4R,5S,6R)-3-amino-2-[(1R,2R,3S,4R,6S)-4,6-diamino-2,3-dihydroxy-cyclohexyl]oxy-5-hydroxy-6-(hydroxymethyl)oxan-4-yl]oxymethoxy]-6-(hydroxymethyl)oxane-2,5-diol		-16.16	-8.32	-9.37
2OE8	Apramycin		-19.57	-8.96	-9.27
6E8T	1-methyl-4-{[3-(2-oxopropyl)-1,3-benzothiazol-3-ium-2-yl]methyl}quinolin-1-ium		-18.64	-10.99	-10.48
3Q3Z	Cyclic diguanosine monophosphate		-24.79	-11.43	-11.81
3GX3	S-adenosyl-l-homocysteine		-15.14	-8.14	-8.56
4LX6	2-aminopyrimido[4,5-d]pyrimidin-4(3H)-one		-15.23	-9.52	-10.68

Supplementary Table S2. Stability metrics of RNALig and MM-PBSA binding free energy predictions. Mean binding free energies (ΔG), standard deviations (SD), and coefficients of variation (CV) are shown for representative RNA–ligand complexes. Lower SD and CV values indicate greater numerical stability of RNALig predictions compared to MM-PBSA.

PDB	Method	Mean_ΔG (Mean binding free energy)	SD (Standard Deviations)	CV (Coefficients of Variation)
2O3W	RNALig	-8.435	0.084	0.01
	MMPBSA	-20.889	5.836	0.279
1F1T	RNALig	-8.799	0.062	0.007
	MM/PBSA	-18.92	4.693	0.248
4TS0	RNALig	-8.671	0.094	0.011
	MM/PBSA	-29.624	3.853	0.13
4TS2	RNALig	-8.551	0.112	0.013
	MM/PBSA	-28.974	4.392	0.152
4ZNP	RNALig	-5.265	0.419	0.08
	MM/PBSA	-15.598	4.141	0.265